

Mini Project 2: Financial Ratios and DuPont Analysis

Louis Zhang

March 2026

1 Question 1: Permanent Library and Final Dataset

I created a permanent SAS library `C` in my WRDS cloud home space and saved the final cleaned and winsorized dataset as `C.final.LZ`. This satisfies the requirement that the final dataset in Step 4 be stored in a permanent library.

2 Question 2: Sample Construction and Variable Design

I used `Comp.Funda` and constructed a firm-year sample covering fiscal years 2018 to 2022. The sample was restricted to standard industrial, domestic, and consolidated observations, using the usual Compustat filters `indfmt='INDL'`, `datafmt='STD'`, `popsrc='D'`, and `consol='C'`. I further required positive total assets, sales, and shareholders' equity so that the financial ratios were economically meaningful.

The resulting sample contains 31,360 firm-year observations and 8,106 distinct firms. For each observation, I retained `gvkey`, firm name, fiscal year, historical SIC code, and the accounting variables needed to construct DuPont ratios. I also created a two-digit SIC code and additional firm characteristics that are likely to affect ROE, including liquidity, solvency, market-to-book ratio, and firm size.

The three-factor DuPont decomposition used in this project is

$$ROE = \text{Profit Margin} \times \text{Total Asset Turnover} \times \text{Equity Multiplier}.$$

The variables are defined as follows:

- $ROE = \frac{ni}{seq}$
- $\text{Profit Margin (PM)} = \frac{ni}{sale}$
- $\text{Total Asset Turnover (TAT)} = \frac{sale}{at}$
- $\text{Equity Multiplier (EM)} = \frac{at}{seq}$

- Current Ratio (CR) = $\frac{act}{lct}$
- Debt-to-Equity (DE) = $\frac{lt}{seq}$
- Market-to-Book (MTB) = $\frac{prcc.f \times csho}{seq}$
- Size = $\log(at)$

3 Question 3: Winsorization, Summary Statistics, and Interpretation

To reduce the influence of extreme outliers, I winsorized the continuous variables at the 1st and 99th percentiles. Tables 1 and 2 summarize the variables before and after winsorization.

Table 1: Selected Summary Statistics Before Winsorization

Variable	N	Mean	Median	99th Pctl
ROE	31,360	-1.4605	0.0570	1.3839
PM	31,360	-14.0708	0.0416	1.4461
TAT	31,360	16.9993	0.4143	3.0786
EM	31,360	6.5368	2.3831	33.9270
CR	24,811	5.6003	1.8274	27.9006
DE	31,302	5.4522	1.3508	32.6632
MTB	28,568	104.5968	1.7565	65.0323
Size	31,360	6.8616	6.9890	12.8632

The raw sample contains very large extreme values, especially for ROE, PM, TAT, EM, and MTB.

Table 2: Selected Summary Statistics After Winsorization

Variable	N	Mean	Median	99th Pctl / Max
ROE	31,360	-0.1966	0.0570	1.3839
PM	31,360	-2.6733	0.0416	1.4461
TAT	31,360	0.5884	0.4143	3.0786
EM	31,360	4.2243	2.3831	33.9270
CR	24,811	3.1506	1.8274	27.9006
DE	31,302	3.1704	1.3508	32.6632
MTB	28,568	4.2482	1.7565	65.0323
Size	31,360	6.8620	6.9890	12.8632

After winsorization, the maximum values are effectively capped at the 99th percentile, which shows that outliers were properly controlled.

The evidence indicates that the data were properly winsorized. Before winsorization, the sample contained extremely large outliers, such as a maximum ROE of 1000, a maximum PM of 2033, a maximum TAT of 514000, a maximum EM of 29586, and a maximum MTB above 2.2 million. After winsorization, the lower and upper extremes were capped at the 1st and 99th percentiles, while the medians remained unchanged. This is exactly what should happen under winsorization: extreme values are trimmed back without materially altering the center of the distribution.

Table 3: Mean and Median of ROE and DuPont Components After Winsorization

Variable	Mean	Median
ROE	-0.1966	0.0570
PM	-2.6733	0.0416
TAT	0.5884	0.4143
EM	4.2243	2.3831

Table 3 shows that the mean values of ROE and PM are much lower than the medians, which indicates strong left-skewness driven by loss-making firms. By contrast, TAT is positive and relatively stable, while EM is substantially above 1 on average, suggesting that leverage plays an important role in explaining cross-sectional variation in ROE.

4 Question 4: Trends in the DuPont Three Factors Over 2018–2022

The yearly mean and median values of ROE and the three DuPont components are shown in Table 4. Figures 1–3 plot the three factors over the five-year period.

Table 4: Yearly Mean and Median Values of ROE and DuPont Components

Year	ROE Mean	PM Mean	TAT Mean	EM Mean	ROE Median	PM Median	TAT Median	EM Me
2018	-0.1577	-2.1751	0.6443	4.0295	0.0656	0.0475	0.4429	2.
2019	-0.2144	-2.3449	0.6005	4.2432	0.0551	0.0394	0.4307	2.
2020	-0.2404	-2.8067	0.5357	4.3218	0.0289	0.0243	0.3709	2.
2021	-0.1342	-2.9767	0.5498	4.0792	0.0718	0.0536	0.3810	2.
2022	-0.2380	-3.0398	0.6130	4.4483	0.0618	0.0425	0.4447	2.

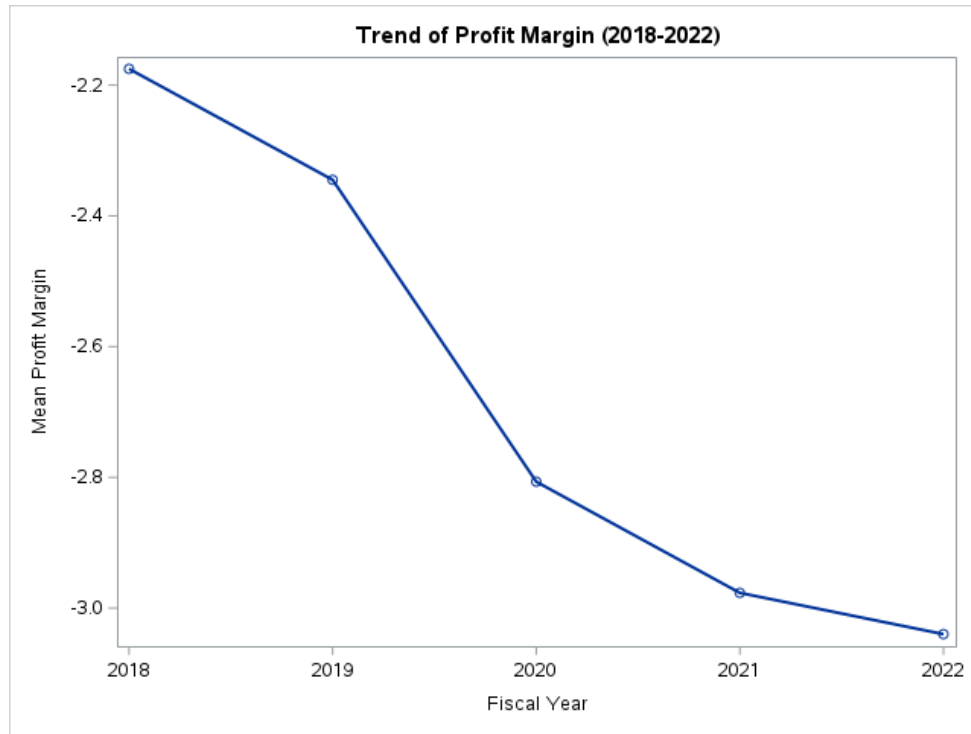


Figure 1: Trend of Profit Margin (2018–2022)

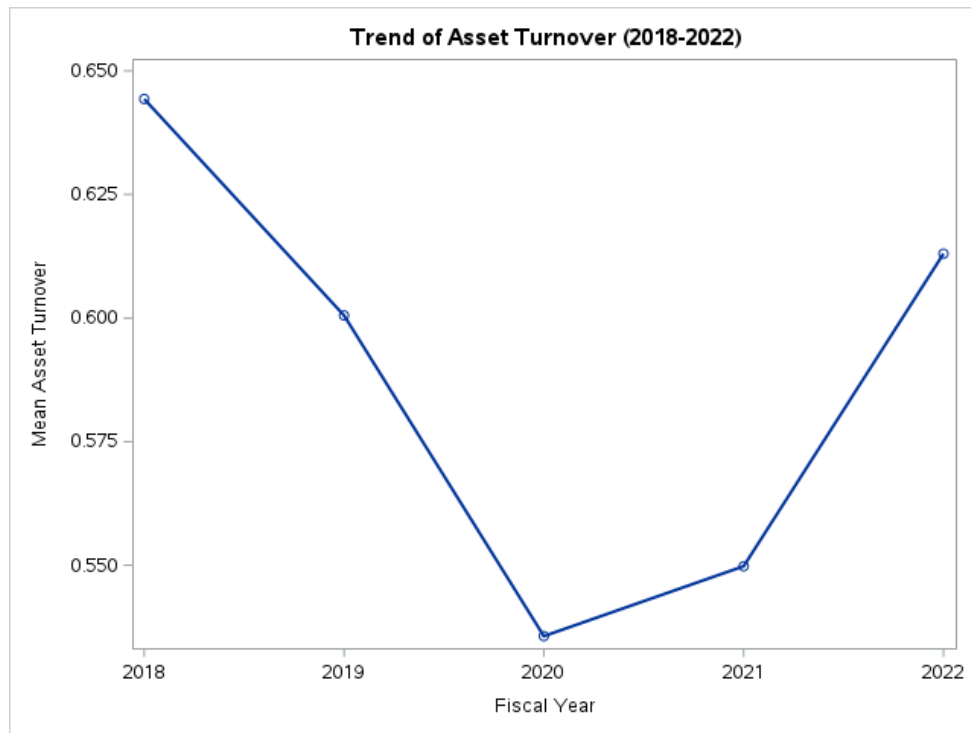


Figure 2: Trend of Asset Turnover (2018–2022)

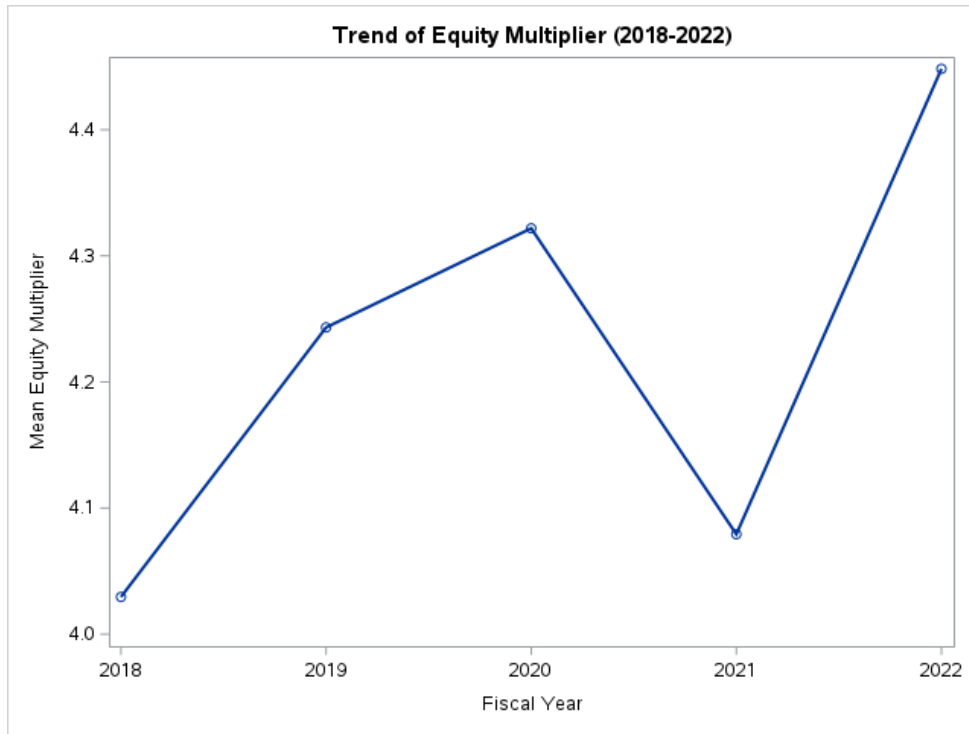


Figure 3: Trend of Equity Multiplier (2018–2022)

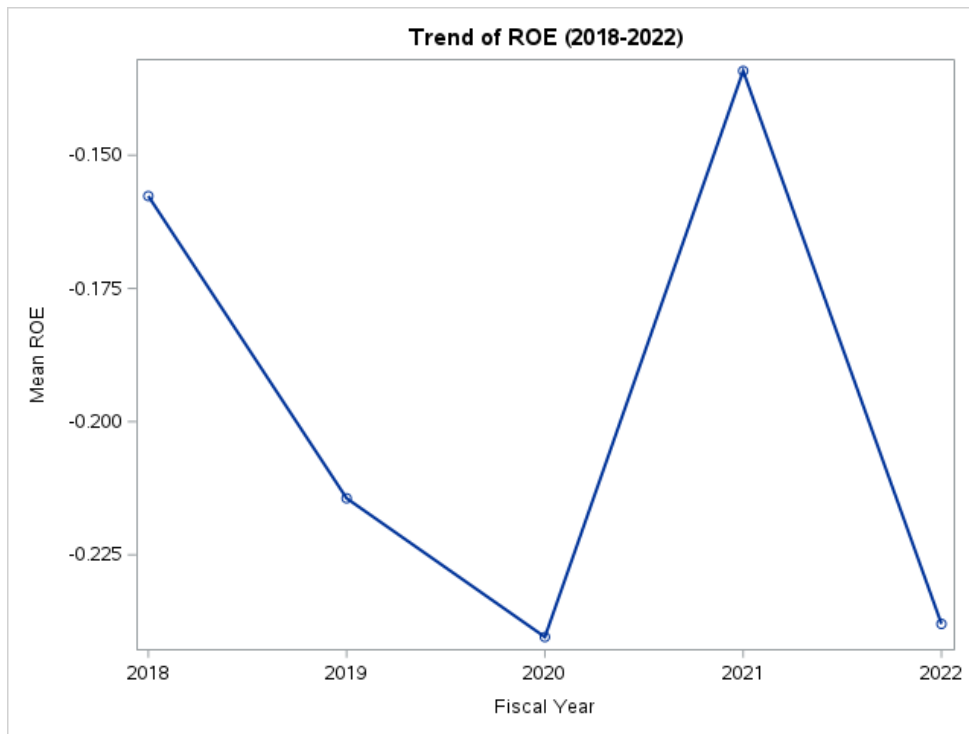


Figure 4: Supporting Plot: Trend of Mean ROE (2018–2022)

Several conclusions follow from these figures. First, profit margin becomes steadily more negative

from 2018 to 2022, which indicates a deterioration in average profitability across the sample. Second, asset turnover declines sharply in 2020 and then partially recovers in 2021 and 2022, suggesting that operating efficiency weakened during the shock period and later improved. Third, the equity multiplier remains elevated throughout the sample and rises further by 2022, implying that leverage intensified over time.

Taken together, these results suggest that the weakness in ROE is primarily associated with poor profitability, while changes in efficiency and leverage partly offset one another. The rebound in asset turnover after 2020 is not strong enough to fully compensate for the increasingly negative profit margin, and the higher equity multiplier in 2022 points to greater reliance on leverage rather than stronger underlying performance.

5 Optional Supporting Analysis: Regression Evidence

Although regression analysis was not explicitly required, I also estimated a cross-sectional regression of ROE on the DuPont factors and firm characteristics. The model uses 22,321 observations after excluding rows with missing values. The overall model is statistically significant, with an F -statistic of 1607.00 and an R^2 of 0.3352.

Table 5: Regression of ROE on DuPont Factors and Firm Characteristics

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	-1.5003	0.0276	-54.34	< 0.0001
PM	0.0122	0.0004	30.71	< 0.0001
TAT	0.2819	0.0105	26.84	< 0.0001
EM	0.1192	0.0151	7.90	< 0.0001
CR	0.0337	0.0016	21.36	< 0.0001
DE	-0.2140	0.0154	-13.85	< 0.0001
MTB	-0.0073	0.0008	-9.45	< 0.0001
Size	0.1659	0.0027	62.35	< 0.0001

The regression results are broadly consistent with expectations: higher profit margin, asset turnover, and equity multiplier are all positively associated with ROE, while greater leverage measured by debt-to-equity is negatively associated with ROE in this specification. The positive coefficient on size suggests that larger firms tend to have stronger ROE after controlling for other variables.

6 Question 5: Clarity, Readability, and Professionalism

The report presents all required components in a clear format: a well-defined sample description, explicit ratio definitions, before-and-after winsorization evidence, summary tables for ROE and the DuPont factors, yearly trend plots, and a written conclusion. The figures use separate panels and

labeled axes to improve readability, and the tables are formatted in a professional style suitable for submission.